



VERIFICATION REPORT FOR “TAIBA N’DIAYE WIND POWER”



Document Prepared by Earthood Services Private Limited

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Summary:

Earthood Services Private limited (hereafter referred to as ESPL) has been contracted by Aera Group to conduct the verification of the project – “Taiba N’Diaye Wind Power”, (VCS ID-2588) with regards to the relevant requirements of VCS programme guidelines and standard (VCS standard version 4.3)/05/ & VCS program guide version 4.2/06/). Relevant requirements of the UNFCCC for CDM project activities/18/, as well as criteria for consistent project operations, monitoring and reporting has been applied for verification.

The monitoring period covered under this verification are from 01/08/2021 to 31/07/2022(inclusive of both dates).

The verification includes confirming the implementation of the monitoring plan of the registered VCS PD (VCS ID 2588)/01/ and the application of the monitored methodology is as per ACM0002- “Grid-connected electricity generation from renewable sources” (Version-20.0)/08/

The project activity involves electricity generation by wind power project of 158.7MW wind farm in Senegal and supplying the electricity to the regional electricity grid. The project being a renewable energy generation activity results in reduction of greenhouse gas (GHG) emissions that are real, measurable, and verifiable and also plays beneficial role in the mitigation of climate change.

A risk-based approach has been followed to perform this verification. During verification 03 Corrective Action request (CARs), 00 Forward Action request (FARs) and 02 Clarification request (CLs) were raised and successfully closed.

The review of the project description, monitoring report, and additional documents related to baseline and monitoring methodology; the subsequent background investigation, on-site interviews and stakeholders have provided ESPL with sufficient evidence to validate the fulfillment of the stated criteria. ESPL confirms that the project is implemented in accordance with the registered VCS PD/01/. The monitoring system is in place and the emission reductions are calculated without material misstatements. Our opinion relates to the project’s GHG emissions, and the resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring and its associated documents. Based on the information seen and evaluated we confirm that the emission reductions from the project activity, ‘Taiba N’Diaye Wind Power’ during the period 01/08/2021 to 31/07/2022 amounts to 226,249 tons of CO_{2e} with displacing 394,851 MWh of electricity from fossil fuel dominating energy grid with electricity generated using wind energy system.

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1 INTRODUCTION

1.1 Objective

Earthood Services Private Limited (ESPL) has been contracted by Aera Group, to undertake the verification of the renewable energy project titled “Taiba N'Diaye Wind Power” (VCS ID-2588). The verifiers have reviewed the GHG data collected to date for the monitoring period from 01/08/2021 to 31/07/2022 covered in this verification.

The objective of this verification is a thorough and independent assessment of registered project activities against the applicable VCS requirements by the VVB. The verification process shall determine whether the proposed project activity complies with the requirement of latest VCS guidelines, applicability conditions of the selected methodology, relevant host country regulations and guidance issued by the VCS Board.

1.2 Scope and Criteria

The scope of this verification is to assess the claims and assumptions made in the VCS monitoring report/02/ against the VCS criteria, including but not limited to, VCS standard, V4.3/05/, applied methodology/08/ and other relevant rules and requirements established for VCS project activities.

The verification is not meant to provide any consulting towards the project participants. However, stated requested for clarification and/or correction actions requested may have provided inputs for improvement of the project design.

1.3 Level of Assurance

☒ Reasonable level of assurance

☐ limited level of assurance

The level of assurance of the verification report falls under reasonable assurance engagement as selected by the client. Reasonable assurance is a high level of assurance regarding material misstatement, but not an absolute one.

Reasonable assurance includes the understanding that there is a remote likelihood that material misstatements will not be prevented or detected on a timely basis. To achieve reasonable assurance, the auditor needs to obtain sufficient appropriate audit evidence to reduce audit risk to an acceptably low level. This means that there is some uncertainty arising from the use of sampling, since it is possible that a material misstatement will be missed.

The evidence used to achieve a reasonable level of assurance is specified in section 2.3 and 2.4 of this report.

1.4 Summary Description of the Project

The project activity is wind power project of 158.7 MW developed by PETN in Senegal. The project activity consists of 46 Vestas wind turbines each totalling the total generation capacity to 158.7 MW.

The aim of the project is to generate cleaner electricity using wind power project and displacing the amount of electricity generated using the grid connected power plant. The electricity generated is supplied to SENELEC, the state-owned utility.

The remote site audit for the project activity was conducted on 24/11/2022 and it was checked that the current monitoring period from 01/08/2021 to 31/07/2022 involves supply of 394,851MWh of electricity to the grid and the total emission reductions of 226,249 tCO₂e.

The start date of the project activity is 09/12/2019 which is date of commissioning of the PETN first tranche under the project activity. The crediting period of the project is from 09/12/2019 to 08/12/2029, the same is verified using the registered PD/01/, previous validation and verification report/04/ and the remote-site audit conducted on 24/11/2022/10/.

The project has applied methodology ACM0002, version 20.0, "Grid- connected electricity generation from renewable sources/08/. The project activity is undergoing second verification and description of the verification was also verified in the previous verification report/04/

Based on the assessment of the documents, and the remote-site audit, the assessment team is able to confirm that there is major shut down due to blade repairs and main bearing issues in this monitoring period are as follows:

Blade Repairs:

Turbines	Stop Date	Start Date
E20	09/21	10/21
E42	01/21	11/21
E40	04/21	09/21
E33	12/20	09/21

Main bearing issues:

Turbines	Stop date	Start Date
E06	07/01/2022	17/04/2022
E38	15/03/2022	04/06/2022
E39	11/03/2022	13/07/2022

2 VERIFICATION PROCESS

The registered VCS project is undergoing second verification, the approach adopted to ensure the quality of emissions reductions is described in the following sections.

2.1 Method and Criteria

The verification approach consists of two phases:

In the first phase, ESPL completed a strategic review and risk assessment of the projects activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leaking if relevant.
- Protocols used to estimate or measure GHG emissions from these sources.
- Collection and handling of data.
- Controls on the collection and handling of data.
- Means of verifying reported data.
- Compilation of the verification Report.

At the end of this phase, ESPL produced a Verification Checklist which, based on the risk assessment of the parameters and data collection and handling processes for each of those parameters, describes the verification approach and the sampling plan. In the second phase, ESPL verified the implementation of the monitoring plan and the data presented in VCS MR/02/ for the period in question. This involves remote-site interviews of project proponent representative's and a desk review of the Monitoring Report. This verification report describes the findings of this assessment.

2.2 Document Review

The verification is performed primarily as a document review of the registered VCS PD/01/ & MR/02/, previous verification report/04/. The review is performed by assessment team using dedicated protocols. The assessment team cross checks the information provided in the documents and information from source other than those used, if available, and also conducts independent background investigations. Earthood conducts a desk review as under:

- A review of the data and information presented to verify their completeness;

- A review of the monitoring plan, the monitoring methodology/08/ including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures.

An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emissions reduction.

2.3 Interviews

The VVB conducted a remote-site audit was conducted on 24/11/2022 i.e., the representatives of the PP were interviewed remote-site/10/, personnel responsible for monitoring of the project activity, data collection and management, and QA/QC procedure. The details of the people interviewed are mentioned in the table below:

S. No.	Name	Organization	Topic covered	Team Member
1.	Mr. Alexandre Dunod	Aera Group (Carbon advisor and offtaker)	Project implementation, Commissioning dates as per the requirements of VCS, implementation status	Shreya Garg, Ashish Yadav
2.	Mr. Amadou Sow	Parc Eolien Taiba N'Diaye S.A.U (PETN) (ESG Manager)	Reliability & accuracy of reading considered for emission reduction calculations, calibration procedure Monitoring and measuring system. Collection of measurements Observations of established practices for verification of monitoring parameters Ongoing Local stakeholders' consultation/19/ process	Shreya Garg, Ashish Yadav

3.	Mr. Yassine Majdallah	Parc Eolien Taiba N'Diaye S.A.U (PETN) (Plant Manager)	Electricity generation records (monthly energy statements, Invoices) Emission reduction calculation	Shreya Garg, Ashish Yadav
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During the remote-site visit, the PP representatives were questioned about the implementation of the project activity, commissioning dates of meter, the generation, recording, and monitoring of the data and the error accountability were discussed. Various documents like JMR sheet, invoices slip etc. were also verified from the reference documents.

Several photographs were also clicked at the site to capture the name plate data, meter specifications, key technical specification of the major equipment like WTG, inverter and meter to establish the current status and implementation of the project activity.

Interview with local stakeholders was also conducted during the remote-site audit on 24/11/2022, the details of the same can be found in the table below:

S.No.	Name	Affiliation	Topic Discussed	Team Member
1.	Fatou	Taiba N'Diaye women association representative(Local Stakeholder)	Grievance Mechanism, Project implementation, Employment opportunities	Shreya Garg, Ashish Yadav
2.	Cheikh Tidiane Keita	Headmaster of Taiba High School (Local stakeholder)	Grievance Mechanism, Project implementation, Employment opportunities	Shreya Garg, Ashish Yadav
3.	Ablaye Fall	General secretary of Muncipal council	Grievance Mechanism, Project implementation, Employment opportunities	Shreya Garg, Ashish Yadav

It was concluded that the local people were aware of the grievance register and there were no negative comments from local stakeholder's side. Instead, the local people appreciated the development and job created by the project activity in that area. This was confirmed through the local expert with whom the recordings of entire meeting were shared. The local expert reviewed and confirmed all the information's received. Thus, a reasonable level of assurance was met.

2.4 Site Visits

As discussed in the above section, remote site inspection is done for the current verification. Following are the details that were verified during the remote- site visit of the project:

Remote- site audit Date – 24/11/2022			
S.no	Activity Performed	Site Location	Team Member
1.	Assessment team checked the implementation of the project. Baseline emission, Emission reduction calculation, Technical description of the project and monitoring. Assessment team also checked that weather the monitoring plan as described in PD is actually practised on -site.	Commune of Taiba N'Diaye, Senegal.	Shreya Garg, Ashish Yadav

A site visit was undertaken by the assessment team on 24/11/2022 to carry out the following:

- An assessment of the implementation and operation of the registered project activity as per registered PD or any approved revised PD/01/ and MR/02/;
- A review of information flow of generation, aggregating and reporting the monitoring parameter.
- Interview with relevant personnel to determine whether the operational and data collection procedure is implemented in accordance with the monitoring plan in the PD.
- A cross- check between information provided in the monitoring report and data from other sources such as JMR sheet, invoices slips, PPA agreement etc;/09,19/
- A check of the monitoring equipment including calibration performance and observation of monitoring practices against the requirement of the PD/01/, the applied methodology/08/ including applicable tool/14,15/ and where applicable, the applied standardized baseline.
- A review of calculation and assumptions made in determining the GHG data and emission reductions.

- An identification of quality control and quality assurance procedure in place to prevent or identify and correct any errors or emission in the reported monitoring parameters.

It is noteworthy that no sampling plan based on the above assessment it can be concluded that the assessment team has verified sufficient appropriate audit evidence, to reduce audit risk to an acceptably low level as requisite to achieve reasonable level of assurance for the current verification.

2.5 Resolution of Findings

The objective of the step is to identify, discuss, and conclude on the issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions. This is done based on the desk review and remote-site assessment/10/. The verification team prepares and/or updates a verification protocol (internal document) that records the conformities and nonconformities, which may be of following types:

CAR (Corrective Action Request) is raised if one of the following occurs:

- a) Non-compliance with the monitoring plan, the methodology/08/ or the standardized baseline are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient.
- b) Modifications to the implementation, operation and monitoring of the registered project activity/01/ has not been sufficiently documented by the project participants.
- c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions.
- d) Changes to the key sustainable development indicators.
- e) Issues identified in a FAR during validation to be verified during verification or previous verification(s)/04/ have not been resolved by the project participants.

FAR (Forward Action Request) is raised during verification if the monitoring and reporting require attention and/or adjustment for the next verification period.

The total of 02 CLs and 03 CAR were raised in the current verification. No FAR was raised. All the findings that are raised and communicated to project participant during the verification are included under Appendix 3. The section also includes the responses by the project participants and an assessment by the verification team if it was closed out or otherwise.

2.5.1 Forward Action Requests

The project activity is undergoing second verification. No FAR is raised in the current monitoring period. There are no open FARs from previous verification or validation/04/ processes which have any impact on current verification.

Details of which can be checked from appendix 3 of this verification report.

2.6 Eligibility for Validation Activities

This part is not applicable for present verification, as ESPL holds the accreditation for the validation of project under this sectoral scope.

3 VALIDATION FINDINGS

Not applicable.

3.1 Participation under Other GHG Programs

The project is registered under VCS. (VCS ID- 2588)

The PP has submitted the declaration/07/ which states that the net GHG emission reductions generated by the project activity will not be used for compliance with any other emissions trading program or to meet binding limits on GHG emissions for the same monitoring period.

CL#01 was raised regarding the same and closed as the PP submitted the declaration/07/.

3.2 Methodology Deviations

There is no deviation in the methodology during the current monitoring period.

3.3 Project Description Deviations

There is no project description deviation during the current monitoring period.

3.4 Grouped Project

This is not applicable, as this is not a grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

The project activity involves installation of 46 Vestas wind turbines to generate electricity from wind. The installed capacity of the project is 158.7 MW in the commune of Taiba N'Diaye, Senegal. Since, this is the second verification, the commissioning, capacity, project implementation, grid connectivity, and technical specifications have been checked from the registered documents and previous verification report/04/.

This proposed wind power project will reduce the GHG emissions generated by the current generation energy mix in SENELEC Power Grid, which is dominated by fossil fuel-based grid connected power plants. The Power generated through the proposed project activity will be supplied to the state utility under a long-term power purchase agreement, as verified using the registered PD/01/ and previous verification report/04/.

The technical specification of the project activity equipment's was verified during the remote-site audit as well as manufacturer technical specifications document provided by PP and all the equipment's installed on-site were found to be consistent with the details mentioned under monitoring report/02/. The current status of the project activity is verified through generation data for WTGS and hence it is confirmed that the project is fully functioning.

The assessment team has reviewed the monthly share certificates/09/, previous verification report/04/ and confirmed that the WTGs are connected to SENELEC substation. Hence, in order to calculate the net electricity exported to the grid by the WTGs of the project activity, the state electricity utility uses an apportioning procedure that has been correctly described in section 4.3 of the MR/02/ and in the PD/01/. This was also verified by interviewing the staff on site during the on-site audit.

It is verified that share certificates are issued by state utility/09/ and the apportioning procedure is carried out by the state utility and the PP has no role in this calculation. The required monitoring system have been installed and are operational. The meters comply with appropriate quality standards applicable for the technology. The accuracy class of the meters installed for the project activity was verified against the registered VCS PD/01/ and VCS MR/02/.

Monitoring Procedure:

The monitoring of the data takes place at metering point and main meter at the SENELEC substation. Meter reading is taken by the state utility on the monthly basis and accordingly monthly share certificates are prepared.

Net electricity supplied to the grid by the project developer is calculated by the state utility using appropriate procedure, adjusting the transmission loss. The same was also verified during remote site audit with site personal deployed at respective sites/10/. The values were cross-checked with the invoices provided by the PP as supporting document.

Values of the parameter “Quantity of net electricity supplied to the grid in year y” i.e., $EG_{\text{Facility } y}$ is directly sourced from the monthly share certificates issued by SENELEC/09/ that indicates the share of electricity for project activity received at the respective pooling sub-stations. The measurement results are cross checked with records of share certificates/09/ and it is in line with applied methodology. Thus, this parameter is considered for the emission reduction calculations/03/.

The net electricity supplied to the grid by project activity is being measured continuously by energy meters located at the pooling substations and recorded on monthly basis. This is in line with the methodology and is accepted.

Assessment team concludes the following:

- There are no material discrepancies between project implementation and the project description provided in the registered PD/01/
- The monitoring plan is implemented completely and monitoring system (i.e., process and schedule for obtaining, recording, compiling and analysing the monitored data and parameters) is appropriate.
- There are no material discrepancies between the actual monitoring system, and the monitoring plan set out in the project description and the applied methodology/08/.
- The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading/07/.

In the view of the information as verified above the assessment team is able to conclude that the project has been implemented as described in the project description.

4.2 Safeguards

4.2.1 No Net Harm

The project activity does not involve any negative environmental or socio-economic impacts.

There is no negative impact to any socio-economic conditions of the region due to the project activity. This project activity does not involve any negative environmental or socio-economic impacts, as the project activity involves generation of power using wind energy which is a clean source of energy. Hence no mitigation measures are required.

4.2.2 Local Stakeholder Consultation

The project activity is undergoing second verification and the local stakeholder's consultation/18/ was appropriately conducted prior to validation as a way to inform the design of the project and maximize participation from stakeholders.

The details of the local stakeholder consultation/19/, along with the concerns raised can be found in the section 2.2 of the MR.

The project proponent has also established a grievance register for continuous stakeholder interactions have been provided to the local stakeholders to lodge their complaints, the detail description of the same can be found in section 2.2 of MR.

No finding were raised regarding ongoing local stakeholder consultation/19/.

4.3 AFOLU-Specific Safeguards

Since the project is non-AFLOU project, hence not applicable.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

The calculations for the emission reductions are found to be correct. The details of the reported and the verified values of all the parameter are listed in section 4.5 of this report.

This data has been verified as described in section 4.5 below. The formulae & method used to calculate the baseline emissions, project emissions and leakage are appropriate and in line with the approved methodology ACM0002 version 20.0/08/.

The PP has calculated the grid emission factor as per the combined margin approach described in the 'Tool to calculate the emission factor for an electricity system', version 2.2.1/14/.

Formula used to calculate Emission reduction:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where,

BE_y = Baseline Emissions in year y (t CO_{2e}/yr)

$EG_{PJ,y}$ = Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = CO₂ emission factor of the grid in year y (t CO₂/yr)

Since the project activity is the installation of a new grid connected renewable power plant the $EG_{PJ,y}$ is calculated as :

$$EG_{PJ,y} = EG_{facility,y}$$

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$$EF_{grid,CM,y} = 0.573tCO_2/MWh$$

The assessment team is able to confirm that the GHG emission reductions and removals have been quantified correctly in accordance with the project description and applied methodology/08/.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All the data recorded are in compliance with the registered VCS PD/01/. The assessment team has checked the monthly share certificates for electricity generated and supplied by project activity/04/ for the current monitoring period to verify the values of monitoring parameters reported in ER calculation sheet/03/ and found to be consistent. Since certificates are prepared and issued by state utility based on the monthly joint meter readings by PP and also used for commercial invoice purpose, they are found to be reliable and authentic.

The monitoring of the project activity is found to be in accordance with the monitoring methodology described in ACM0002, Version 20.0/08/. The monitoring mechanism is effective and reliable. During the Remote-site audit, personnel involved at various levels of the operation of the project activity have been interviewed to confirm that the plant personnel are conscious of the importance of the monitoring activities.

The meters comply with appropriate quality standards applicable for the used technology. The accuracy class of the meters installed for the project activity was verified through the registered VCS PD/01/· MR/02/, and calibration certificates/11/ are found to be consistent.

The following parameter has been verified for current monitoring period:

Parameter: Quantity of net electricity generation supplied by the project plant/unit to the grid in year y, $EG_{\text{facility},y}$ (MWh):

Means of Verification	Criteria/ Requirements	Assessment/ Observation
	Measuring/Reading/Recording frequency	This parameter is measured directly with electricity meters at substation and reported directly Senelec by PETN on monthly basis.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The measuring and reporting frequency is in line with the monitoring plan as outlined in the registered PD/01/ and monitoring methodology/08/.
	Monitoring equipment	The meter off 0.2s accuracy class is used. (calibration details of the meter is provided separately in this section under

		the heading “calibration of meters”)
	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer’s specification?	All the energy meters are of accuracy class 0.2s. Accuracy of the monitoring equipment’s is in accordance with the monitoring plan as outlined in the registered PD. /01/
	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	Yes, the accuracy of the monitoring equipment is valid for entire monitoring range.
	Calibration frequency /interval:	The meters are calibrated every year which is done in line with the procedures mentioned in the PD, however during this monitoring period a delay in calibration was observed of 2 months at SENELEC substation. The PP has applied an error factor of 0.2% on the value where delay in calibration was observed.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed or as per the manufacturer’s specifications?	The meters are calibrated every year which is done in line with the procedures mentioned in the PD, however during this monitoring period a delay in calibration was observed of 2 months at SENELEC substation. The PP has applied an error factor of 0.2% on the value where delay in calibration was observed.

	Is the calibration of measuring equipment carried out by an accredited person or institution?	The meter testing and checking procedures are solely within the purview of SENELEC (PETN).
	How the values in the monitoring report verified?	The value in the monitoring report were verified from the monthly JMR and share certificates/09/. Value monitored: 394,851 MWh
	Is(are) calibration(s) valid for the whole reporting period?	Yes, the calibration is valid for the entire monitoring period The PP has applied an error factor of 0.2% on the value where 2 months delay in calibration was observed.
	If applicable, has the reported data been cross-checked with other available data?	The monthly reported values of EGy were further cross-checked with share certificates issued by SENELEC (PETN). /09/.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data ensure correct transfer of data and reporting of emission reduction management QA/QC processes are in Place.
	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by Appendix 1 to the CDM Project Standard?	There was a delay in calibration observed during the current monitoring period The meters are calibrated every year which is done in line with the procedures mentioned in the PD, however during this monitoring period a delay in calibration was observed of 2 months at SENELEC substation. The PP has applied an error factor of 0.2% on the

		value where delay in calibration was observed.
Findings	CL#02 and CAR#02 is raised and resolved.	
Conclusion	The parameter has been monitored appropriately in accordance with the registered monitoring plan/01/ and applied methodology/08/. The monitored data was recorded consistently as per the approved frequency in monitoring plan. Since 100% data has been monitored and verified, the verification team can ascertain that the values used for calculation of emission reduction are free from material errors.	

Parameters fixed ex ante:-

EF_{grid,CM,y}: The parameter is described as combined margin CO2 emission factor of Electricity Grid which is measured in units tCO₂e/MWh. The value provided in MR i.e., 0.573 has been verified from PD/01/.

Calibration of meters:

The energy meters installed at sub-station is of accuracy class 0.2s as verified through the calibration certificates/11/.

The installation of the meters was checked through Calibration certificates. Details of the meters are provided below table:

Substation meter details:

Meter Serial No.	Sub-Station	Make and Accuracy class	Date of previous calibration	Date of latest calibration	Remark
84406426	SENELEC	Itron and Secure 0.2s	02/10/2020, valid until 02/10/2021	09/12/2021	There is delay in calibration from 02/10/2021 to 09/12/2021 and taking a conservative approach an error factor of 0.2% is applied on the value of net electricity applied to the grid (EGfacility,y = Exports (1-0.2%) – Imports (1+0.2%))
84406427	SENELEC	Itron and Secure 0.2s	02/10/2020, valid until 02/10/2021	09/12/2021	There is delay in calibration from 02/10/2021 to 09/12/2021 and taking a conservative approach an error factor of 0.2% is applied on the value of net electricity supplied to the grid (EGfacility,y = Exports (1-0.2%) – Imports (1+0.2%)).

It was observed there was a calibration delay in meters, therefore the PP took a conservative approach, as the error factors were found to be in the permissible limits of 0.2% (0.2s), the PP applied the maximum error factor of $\pm 0.5\%$ on the values of net electricity supplied to the grid.

The assessment team has verified the monthly JMRs/credit notes issued by state utility and confirmed that only the data recorded through main meters is used to calculate net electricity supplied to the grid for all three feeders and consequently for ER calculations.

GHG Calculations:

The emission reduction as per the applied methodology equals the baseline emissions (project emissions and leakage emissions for such project activities is considered zero). The formula provided for the calculation of baseline emissions is

$$ER_y = BE_y - PE_y$$

Where,

ER_y = Baseline Emissions in year y (t CO₂e/yr)

PE_y = Project Emissions in year y (t CO₂e/yr)

For calculation of baseline emissions:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where,

BE_y = Baseline Emissions in year y (t CO₂e/yr)

$EG_{PJ,y}$ = Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = CO₂ emission factor of the grid in year y (t CO₂/yr)

Since the project activity is the installation of a new grid connected renewable power plant the $EG_{PJ,y}$ is calculated as :

$$EG_{PJ,y} = EG_{facility,y}$$

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$$EF_{grid,CM,y} = 0.573 \text{ tCO}_2/\text{MWh}$$

The verification team confirms that appropriate methods and formulae for calculating baseline emissions have been followed. The assumptions, emission factors and default values that were applied in the calculations are justified. The actual emission reduction achieved during the current

monitoring period are 12% lower than the estimated amount of emission reduction as determined in the registered VCS PD/01/ which is due to the low PLF achieved by the project activity during the monitoring period. The generation depends on wind availability and other climatic conditions which is acceptable as it is not in control of project participant.

All the data were made available and have monitored as per required monitoring frequency. The means of verification for the values of parameters, used for baseline emissions calculations is described above.

CAR#02 and CL#02 was raised regarding transparent description of calibration frequency and it was successfully resolved.

4.6 Non-Permanence Risk Analysis

Not applicable for the project activity

5 VERIFICATION OPINION

Earthood Services Private Limited (ESPL) contracted by Aera Group, to perform the independent verification of the emission reductions for the VCS project activity 'Taiba N'Diaye Wind Power' (VCS ID- 2588) in Senegal for the monitoring period 01/08/2021 to 31/07/2022 as reported in monitoring report version 4 dated 20/02/2023./02/ The Aera Group is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emission reduction from the project activity.

It is our responsibility to express an independent verification statement on the reported GHG emission reductions from the project activity. ESPL commenced the verification on the basis of the baseline and monitoring methodology ACM0002 Version 20.0/08/, the monitoring plan contained in the VCS PD/01/ & MR/02/.

ESPL verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. ESPL planned and performed the verification by obtaining evidence and other information and explanations that ESPL considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion the GHG emissions reductions reported for the project activity for the period 01/08/2021 to 31/07/2022 are fairly stated in the Monitoring Report/02/. The GHG emission reductions were calculated correctly based on the approved baseline and monitoring methodology ACM0002 20.0/08/ and VCS standard version 4.3/05/.

Verification period: From [01-08-2021] to [31-07-2022]

The actual emission reductions for this monitoring period are 226,249 tCO₂e. Whereas the estimated emission reductions for this monitoring period are 257,735 tCO₂e. Thus, there is a variation of 12% (negative) in the total ERs.

Verified GHG emission reductions and removals in the above verification period, broken down by calendar year:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-08-2021-31-12-2021	70,971	0	0	70,971
01-01-2022-31-07-2022	155,278	0	0	155,278
Total	226,249	0	0	226,249

Approved By:



Kaviraj Singh

Managing Director

Earthood Services Private Limited

Date: 24/02/2023

Place: Gurgaon, Haryana

APPENDIX I: DOCUMENT REFERENCES

S. No	Title of documents	Version	Date
1.	Registered PD https://registry.terra.org/app/projectDetail/VCS/2588	02	14/01/2022
2.	Monitoring Report	04	20/02/2023
3.	ER sheet	1.1	26/12/2022
4.	Previous Verification Reports for the project activity Available at Terra registry for project activity page: https://registry.terra.org/app/projectDetail/VCS/2588	03	20/01/2022
	Validation Reports for the project activity Available at Terra registry for project activity page: https://registry.terra.org/app/projectDetail/VCS/2588	03	20/01/2022
5.	VCS standard https://terra.org/wp-content/uploads/2022/06/VCS-Standard_v4.3.pdf	4.3	-
6.	VCS program guide https://terra.org/wp-content/uploads/2022/06/VCS-Program-Guide-v4.2.pdf	4.2	-

7.	Letter of declaration from PP regarding not having created or sought any other form of environmental credits for the same period	-	-
8.	Approved Methodology ACM0002 Ver.-20.0 –“Grid-connected renewable electricity generation”. https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG	20.0	-
9.	Monthly JMR and share certificates	-	-
10.	On-site audit and interviews conducted for the project monitoring period	-	12/10/2022
11.	Calibration certificates for all the meters	-	13/06/2021-08/10/2022
12.	Operations shutdown details	-	-
13.	VCS webpage for the project, VCS ID 2588; https://registry.terra.org/app/projectDetail/VCS/2588	-	-
14.	Tool to calculate the emission factor for an electricity system	2.2.1, EB 63	-
15.	Tool for the demonstrations and assessment of additionality	5.2.1, EB 39	-
16.	Baseline carbon dioxide Emission Database by CEA	6	01/03/2011
17.	CDM web page link for project activity, Project 6349 https://cdm.unfccc.int/Projects/DB/DNV-CUK1330446091.13/view	-	-
18.	Stakeholder's consultation	-	22/12/2012-28/12/2012
19.	PPA Agreement	-	-

APPENDIX II: ABBREVIATIONS

Abbreviations	Full texts
ABT	Availability Based Tariff
BEF	Baseline Emission Factor
BM	Build Margin
CAR	Corrective Action Request
CL	Clarification Request
CMS	Central Monitoring System
CMP	Conference of Parties Serving as Meeting of Parties
CO2	Carbon dioxide

DISCOM	Distribution Company
EB	Executive Board
EPC	Engineering, Procurement and Construction
FAR	Forward Action Request
GHG	Green House Gas
ISO	International Standards Organization
JMR	Joint Meter Reading
kW	Kilowatt
kWh	Kilowatt hour
MFR	Multi-Function Relay
MR	Monitoring Report
MW	Megawatt
MWh	Megawatt-hour
O&M	Operation and Maintenance
PD	Project Description
PLF	Plant Load Factor
PP	Project Proponent
PPA	Power Purchase Agreement
QA/QC	Quality Assurance and Quality Control
TOD	Time of the Day
UNFCCC	United Nations Framework Convention on Climate Change
WEC	Wind Energy Convertor
WEG	Wind Energy Generator
WTG	Wind Turbine Generator
VCS	Voluntary Carbon Standard
VCSA	Voluntary Carbon Standard Association
VCS PD	VCS Project Description
VCUs	Voluntary Carbon Units
PETN	Parc Eolien Taiba N'Diaye

APPENDIX III: FINDINGS OVERVIEW

Table 1. Remaining FAR from validation and/or previous verification

FAR ID	XX	Section no.	-	Date : DD/MM/YYYY
Description of FAR				
Project participant response				Date : DD/MM/YYYY

Documentation provided by project participant	
DOE assessment	Date: DD/MM/YYYY

Table 2. CL from this verification

CL ID	01	Section no.	1.9 & 3.1	Date:03/12/2022
Description of CL				
PP is requested to submit the following documents:				
1. Declaration confirming that the GHG Emission reductions or removals generated by the project activity will not be used for compliance with an emission trading program or to meet binding limits on GHG Emissions. 2. Plant shutdown details. 3. Proof of grievances raised during this monitoring period 4. Proof of resolution of grievances raised during this monitoring period				
Project participant response				Date:19/12/2022
Documentation provided by project participant				
1. Declaration by PP for no double counting - Taiba N'Diaye Wind Power – signed 2. 2021-2022 Readings details 3. "PETN_grievance register_20190410 / PETN_ES_2021 annual report" from grievance data file 4. Proof of grievances resolutions				
DOE assessment				Date: 26/12/2022
1. VVB confirms that PP has submitted the signed "declaration for no double counting" to confirm that the GHG Emission reductions or removals generated by the project activity will not be used for compliance with an emission trading program or to meet binding limits on GHG Emissions dated 31/07/2022. Hence, the finding is closed. 2. VVB confirms that PP has provided the shutdown reports of the plant for the year 2021-2022. The evidence is satisfactory and hence the finding is closed. 3. VVB confirms that according to the grievance register, there were six entries and all were closed. Hence, the finding is closed. 4. VVB confirms that the proof provided for grievance resolutions indicate resolved grievances and hence this finding is closed. Hence, CL#01 is closed.				

CL ID	02	Section no.	3.2	Date:03/12/2022
Description of CL				
Under section 3.2, in the description of the parameter 'EGfacility,y' the comment section states, "New Energi Express inspection/testing undertaken on 09/12/2021 as per report, valid until 09/12/2022. Thus, the applicable error of 0,2% was conservatively deducted for the period going from 02/10/2021 to 30/11/2021, based on meters' accuracy (0.2s)."				
PP is requested to clarify in detail that how the correction factor is applied to the project activity.				
Project participant response				Date: 19/12/2022

As per option (a) of paragraph 366 of the “CDM validation and verification standard for project activities” the maximum permissible error is applied “to the measured values taken during the period between the scheduled date of calibration and the actual date of calibration” in case of calibration delay. The permissible error of 0.2% was therefore applied to “net energy generation” of October and November 2021 (see lines 24 and 25 of the “Net EG MWh” column on the ERs calculation excel file).

Documentation provided by project participant

ER Ex-post Calc - TAIBA

DOE assessment

Date: 26/12/2022

VVB confirms that the error factor 0.2% was applied in line with “option (a) of paragraph 366 of the “CDM validation and verification standard for project activities” the maximum permissible error is applied “to the measured values taken during the period between the scheduled date of calibration and the actual date of calibration” in case of calibration delay.” The value has been applied in the ER sheet tab ‘Ex-post’ cells H24 and H25. The evidence is satisfactory and hence, the finding is closed.

Hence, CL#02 is closed.

Table 3. CAR from this verification

Table 6: CAR from this verification

CAR ID	01	Section no.	MR	Date: 03/12/2022
Description of CAR				
Under section 3 of the latest version of MR following have been observed:				
1. The document is not in line with the MR template version 4.1. PP is requested to make the document consistent with the respective MR template.				
2. The page numbers in the MR are not provided.				
Correction is requested.				
Project participant response				Date: 19/12/2022
1. The document is now in line with the MR template version 4.1				
2. Page numbers are now added				
Documentation provided by project participant				
Taiba N'Diaye Wind Power-VCS-MR2 v3				
DOE assessment				Date: 26/12/2022
PP has updated the MR according to the MR template, version 4.1 and the document is now consistent with the respective MR template. The page numbers have also been added.				
Hence, CAR #01 is now closed.				

CAR ID	02	Section no.	3.2	Date:	03/12/2022
Description of CAR					
1. The “Serial ID: 1904406426-1904406427” mentioned under ‘Monitoring equipment’ of parameter EG _{facility,y} , is inconsistent with the calibration certificate provided by the PP. PP is requested to update in accordance with the certificate.					
2. There is an typo error in Parameter EG _{facility} , page no. 19. It is written as 0,2. PP is requested to update.					
Project participant response				Date: 19/12/2022	

1. The Serial ID mentioned under “Monitoring equipment” is now consistent with the calibration certificate provided 2. Corrected	
Documentation provided by project participant	
Taiba N'Diaye Wind Power-VCS-MR2 v3	
DOE assessment	Date: 26/12/2022
PP has updated the Serial ID as per the calibration certificate and the typo error has also been corrected.	
Hence, CAR #02 is now closed.	

CAR ID	03	Section no.	4.3	Date: 03/12/2022
Description of CAR				
Under section 4.3 ‘Monitoring Plan’ of the MR the following have been observed:				
1. The schematic diagram of project activity has not been provided. 2. The details of the meters in the project site are not provided.				
Project participant response				Date: 19/12/2022
1. The schematic diagram is now added under section 4.3 2. Meters details were added				
Documentation provided by project participant				
Taiba N'Diaye Wind Power-VCS-MR2 v3 Schematic diagram				
DOE assessment				Date: 26/12/2022
The PP has provided the schematic diagram and meter details in the updated PD.				
Hence, CAR #03 is now closed.				

Table 4. FAR from this verification

FAR ID	XX	Section No.	-	Date: DD/MM/YYYY
Description of FAR				
Project participant response				Date: DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

APPENDIX IV: COMPETENCY STATEMENTS

Competence Statement			
Name	Shreya Garg		
Country	India		
Education	M.Sc. (Climate Science & Policy), TERI University		
Experience	6 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	AMS.I.A., AMS.I.C., AMS.I.D., AMS.I.F., AMS.II.D., AMS.II.G., AMS.II.J., AMS.III.AV., AMS.III.BL, ACM0002, ACM0012		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2, TA 3.1)		
Reviewed by	Shifali Guleria	Date	26/04/2022
Approved by	Deepika Mahala	Date	26/04/2022

Competence Statement	
Name	Ashish Yadav
Education	M.Sc Environmental Sciences B.Sc Biotechnology
Experience	1 Year
Field	Wastewater treatment
Approved Roles	
Team Leader	NO
Validator	NO
Verifier	NO

Methodology Expert	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Trainee	Yes		
Reviewed by	Shifali Guleria (Quality Manager)	Date	20/09/2022
Approved by	Deepika Mahala (Technical Manager)	Date	20/09/2022

Competence Statement			
Name	Namory Diakhate		
Country	Senegal		
Education	MA in English Studies		
Experience	1+ years		
Field	Renewable energy, Energy Efficiency, Capacity Building		
Approved Roles			
Team Leader	No		
Validator	No		
Verifier	No		
Methodology Expert	No		
Local expert	Yes (Senegal)		
Financial Expert	No		
Technical Reviewer	No		
TA Expert	No		
Reviewed by	Shreya Garg	Date	12/12/2018
Approved by	Anshika Gupta	Date	12/12/2018

Competence Statement	
Name	Deepika Mahala
Country	India
Education	M. Sc. (Environment Management), GGSIP University B.Sc. Hons. (Chemistry), Sri Venkateshwar College, DU
Experience	6 Years +
Field	Climate Change
Approved Roles	

Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	ACM0002, AMS.I.D., AMS.I.A, AMS.III.AV, AMS.II.G, AMS-II.C		
Local expert	YES (India, Bangladesh)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2 & TA 3.1)		
Reviewed by	Shifali Guleria (QM)	Date	28/04/2022
Approved by	Kaviraj Singh (MD)	Date	28/04/2022